Fixture record fx-0009

Title: A model-based systems engineering framework for rural emergency medical response

Abstract:
We present an MBSE framework, implemented in SysML, that models the emergency medical response system for a network of rural clinics coordinating during a regional flood disaster.

Methods
Study design: Model-based systems engineering (MBSE) framework development, illustrated with a SysML case model
Country: United States
Population: A network of rural primary-care clinics coordinating regional flood response
Concept: MBSE framework for structuring emergency medical response coordination
Context: Rural regional flood disaster response

Results
- The SysML-based framework surfaced two previously undocumented coordination gaps between clinics.
- Framework users reported the model improved shared situational awareness in a tabletop exercise.

Limitations
The authors state the framework has not yet been validated against a real flood event and was demonstrated only on a synthetic coordination scenario.

Funding
Funding sources: Regional Rural Health Consortium (fictional, synthetic fixture)
Funder role: The funder had no role in the study.

Synthetic test data — pipeline conformance fixture, not scientific evidence.